



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya College of Engineering



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VIDYAVIHAR UNIVERSITY

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Syllabus

Honours' Degree Programme in Data Science and Analytics

(Offered by Department of Computer Engineering)

From

Academic Year 2024-25

Revision 2

(Approved in Academic Council meeting dated 22/04/2024)



K J Somaiya College of Engineering, Mumbai-77

(Somaiya Vidyavihar University)

Honours' Degree Programme in Data Science and Analytics

Offered by Department of Computer Engineering

Introduction:

In today's data-driven society, Data Science provides a foundation for problem solving that impacts many areas of the economy, including science, engineering, medicine, banking, finance, sports and the arts. Data science is an interdisciplinary field that focuses on analysing large amounts of data in various formats such as text, audio video etc., to extract information from inherent patterns, design and develop underlying models, and use data for real world applications for prediction, detection and decision.

Data processing and analytics converts raw data of different format into meaningful format which can be analysed and interpreted for a variety of purposes. Data Science focuses on data extraction and processing techniques and machine learning algorithms for inferring and representation of data for meaningful presentation for easy interpretation and decision making. Data Science focuses on innovative and intelligent ways of handling data in structured and unstructured format which may be available in large volume, and analyzing data for various purposes.

Data science and analytics programme is designed to prepare the students in interdisciplinary domains for gaining hands-on experience on data analytics, machine learning and data visualization methods as it relates to multiple fields of interest. It is designed to empower the students to employ problem ideation, computational thinking and data science tools to solve practical business problems. The coursework consists of courses that cover the spectrum of Data Science to equip the students with knowledge of data analysis, Machine Learning and data visualization techniques for data-centric computation to address problems of real world.

Objectives:

- Applications of principles of Data Science for analysis of diverse domain problems.

- Use software tools and machine learning algorithms from the areas of statistics, mathematics, Computer Science and Artificial intelligence to model and analyze real-world data, communicate data insights, and effectively present results using data visualization techniques.
- Deployment of latest tools and technologies to analyse large amounts variety of Data.
- Understand ethical practices that are importantly and inevitably tied to data-driven prediction and decision-making.

Learning Outcomes of the Honours' Degree Programme:

At the successful completion of this programme an engineering graduates will be able to

- Understand foundational principles, key concepts, and ethical considerations in data science for recognizing its significance across various interdisciplinary domains.
- Perform data acquisition, cleaning, exploration, and statistical analysis for gaining proficiency in Data Science techniques.
- Apply Machine Learning algorithms to model, develop, and evaluate for solving real-world problems.
- Analyze data to create impactful visualizations for effective communication after assessing the data for recognising visual representation quality.

Assessment Methods: Evaluation is done by a variety of tools including Open book tests, MCQs (multiple choice questions), Study of research papers, Internal Assessment tools and End Semester Examinations etc. Mini-Projects are offered in courses also to encourage project based learning among students.

Acronyms used in syllabus document	
Acronym	Definition
CA	Continuous Assessment
ESE	End Semester Exam
IA	Internal Assessment
O	Oral
P	Practical
P&O	Practical and Oral
TH	Theory
TUT	Tutorial
TW	Term work
ISE	In-semester Examination
CO	Course Outcome

Acronyms used in Course code e.g. 116HxxC301

Position of Digit	Acronym	Definition
1	2	SUV 2023 Second Revision
2	16	KJSCE
3	H	Honour Degree Program
4	03(xx)	Data Science and Analytics
5	C	Core Course
	L	Laboratory Course
	T	Tutorial
	P	Project Based Course
6	1/2/3/4	Semester Number
7	01/02/03--	Course Number

Proposed Credit Scheme

Course Code	Course Name	Teaching Scheme (Hrs.) TH – P – TUT	Total (Hrs.)	Credits Assigned TH – P – TUT	Total Credits	Suggested semester of Honours' degree
216H03C401	Introduction to Data Science	3 – 0 – 0	03	3 – 0 – 0	03	IV
216H03L401	Introduction to Data Science Laboratory	0 – 2 – 0	02	0 – 1 – 0	01	IV
216H03C501	Data Analysis	3 – 0 – 0	03	3 – 0 – 0	03	V
216H03L501	Data Analysis Laboratory	0 – 0 – 2	02	0 – 0 – 1	01	V
216H03C601	Machine Learning	3 – 0 – 0	03	3 – 0 – 0	03	VI
216H03L601	Machine Learning Laboratory	0 – 2 – 0	02	0 – 1 – 0	01	VI
216H03L602	Data Visualization Laboratory	0 – 0 – 2	02	0 – 0 – 2	02	VI
216H03C701	Advanced Data Analytics	3 – 0 – 0	03	3 – 0 – 0	03	VII#
216H03L701	Advanced Data Analytics Laboratory	0 – 2 – 0	02	0 – 1 – 0	01	VII#
	Total	12 – 8 – 2	22	12 – 4 – 1	18	

****All courses are project/mini project based**

#Course will be conducted in semester VIII for students who are opting the semester long internship in semester VII

Proposed Examination Scheme

Course Code	Course Name	Examination Scheme				
		Marks				
		CA		ESE ^s	Lab/ Tut CA	Total
		ISE	IA			
216H03C401	Introduction to Data Science	30	20	50	--	100
216H03L401	Introduction to Data Science Laboratory	-	-	-	50	50
216H03C501	Data Analysis	30	20	50	--	100
216H03L501	Data Analysis Laboratory	-	-	-	50	50
216H03C601	Machine Learning	30	20	50	-	100
216H03L601	Machine Learning Laboratory	-	-	-	50	50
216H03L602	Data Visualization Laboratory	-	-	-	50	50
216H03C701	Advanced Data Analytics	30	20	50	-	100
216H03L701	Advanced Data Analytics Laboratory	-	-	-	50	50
Total		120	80	200	250	650

Course Code	Name of the Course				
216H03C401	Introduction to Data Science				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	03	-	-	03	
Credits Assigned	03	-	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	-	20	30	50	100

Course prerequisites: Elementary mathematics, Basics of programming in Python.

Course Objectives:

1. Understand and Learn methodologies and significance of Data Science ethical considerations while working with various data types and the 5 V's of Data.
2. Comprehend mathematical foundation in probability, statistics, and basic linear algebra for exploratory data analysis.
3. Learn and implement data cleaning, transformation, and feature extraction and selection techniques to prepare knowledge base for data analysis.
4. Apply supervised and unsupervised machine learning techniques for real-world application for prediction, detection and decision.
5. Create effective and engaging data visualizations using visualization tools for presenting, inferring and for communication of insights.

Course Outcomes (CO):

CO 1	Understand the significance and Data Science processes in real world applications.
CO 2	Gain necessary mathematical in probability, statistics, and basic linear algebra.
CO 3	Learn data cleaning, transformation, and feature engineering techniques.
CO 4	Apply supervised and unsupervised machine learning techniques to solve real-world problem.
CO 5	Create compelling data visualizations for finding inferences and effective communication of application outcomes.

Module No.	Unit No.	Contents	No. of Hrs.	CO
1	Introduction to Data Science		08	CO1
	1.1	What is Data Science? Importance and applications of Data Science, Data Science workflow, The 5 V's of Data		
	1.2	Structured, semi-structured, and unstructured data. Challenges and considerations for handling different data types. Ethical Considerations in Data Science.		
2	Mathematical Foundations for Data Science		10	CO2
	2.1	Linear Algebra: Vectors and matrices. Matrix operations and transformations.		
	2.2	Probability Distributions: Gaussian distribution, Binomial distribution, Poisson distribution. Properties of Probability Distributions: Mean, variance, and standard deviation of a distribution. Moments and percentiles. Skewness and kurtosis.		
	2.3	Statistical analysis: Measures of central tendency (mean, median, mode). Measures of dispersion (range, variance, standard deviation). Quartiles, percentiles. Stem and leaf plots, Box plots Hypothesis testing		
3	Exploratory Data Analysis and ETL		09	CO3
	3.1	Data Cleaning and Transformation: Identifying Missing Data, Handling Missing Data, Handling Outliers and Data Types (categorical, numerical, ordinal), and methods for encoding categorical data, including one-hot encoding and label encoding. Importance of data type selection for modeling.		
	3.2	Introduction to Feature Engineering: Define feature engineering and its role in data pre-processing.		
	3.3	ETL Overview		
4	Introduction to Machine Learning		10	CO4
	4.1	Machine Learning Basics: What is machine learning? Types of machine learning: supervised, unsupervised, and reinforcement learning.		
	4.2	Supervised Learning: Linear Regression (simple linear regression, Least Squares Method, Model Evaluation Metrics – MSE & R-squared (R^2) coefficient of determination & Interpretation of R^2 . Naïve Bayes Algorithm.		
	4.3	Various Distance measure. Unsupervised learning – k means clustering.		
		Self-learning: Logistic Regression, R libraries that implement the above algorithms		

5	Data Visualization			
	5.1	Principles of Effective Data Visualization, Case studies of data visualization	08	CO5
	5.2	Tools for Creating Visualizations: Matplotlib, Seaborn		
		Self-Learning : ggplot2 (R), Plotly & Tableau.		
Total			45	--

Reference Books

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/ Year
1	Jiawei Han, Micheline Kamber, Jian Pei	<i>Data Mining: Concepts and Techniques</i>	Morgan Kaufmann	3rd edition, 2012
2	Sheldon M. Ross	<i>Introductory Statistics</i>	Academic Press	4th edition, 2017
3	Avrim Blum, John Hopcroft, and Ravindran Kannan	Foundations of Data Science	ONLINE	2014
4	Jake VanderPlas	<i>Python Data Science Handbook</i>	O'Reilly	2016
5	Cathy O'Neil and Rachel Schutt	<i>Doing Data Science, Straight Talk From The Frontline</i>	O'Reilly	2014, Edition 1

Course Code	Name of the Course				
216H03L401	Introduction to Data Science Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	-	02	-	02	
Credits Assigned	-	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	50	-	-	-	50

Course prerequisites: Elementary mathematics, Basics of programming in Python.

Course Objectives:

1. Understand the significance, workflow, and ethical considerations of Data Science while recognizing various data types and the 5 V's of Data.
2. Develop a solid foundation in probability, statistics, and basic linear algebra for data analysis.
3. Learn data cleaning, transformation, and feature engineering techniques to prepare data for analysis.
4. Apply supervised and unsupervised machine learning techniques to real-world data for prediction and pattern discovery.
5. Create effective and engaging data visualizations using principles and tools for clear communication of insights.

Course Outcomes (CO):

CO 1	Understand the significance and Data Science processes in real world applications.
CO 2	Gain necessary mathematical in probability, statistics, and basic linear algebra.
CO 3	Learn data cleaning, transformation, and feature engineering techniques.
CO 4	Apply supervised and unsupervised machine learning techniques to solve real-world problem.
CO 5	Create compelling data visualizations for finding inferences and effective communication of application outcomes.

Laboratory experiments will be based on the entire syllabus of the course 216H03C401, 'Introduction to Data Science'. Students will be graded based on continuous assessment during laboratory.

Course Code	Name of the Course				
216H03C501	Data Analysis				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites:

- Concepts of DBMS, Probability and statistics, knowledge of programming language (C/C++/Java/ Python).

Course Objectives:

Introduction to the fundamental concepts of Data Analytics , analyse real world case studies by applying mining algorithms and visualization for decision-making in Geospatial, social media ,healthcare and text mining business applications

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1	Understand basic concepts of data analytics to solve real world case studies
CO2	Apply the data analysis on geospatial system
CO3	Perform the graph data analysis
CO4	Perform Time series Analysis
CO5	Apply the text data analytics in the field of Health care

Module No.	Unit No.	Details	Hrs.	CO
1		Introduction to Data Analysis	05	CO1
	1.1	Introduction to Data Analytics, Different types of data analytics: Descriptive analytics, Diagnostics Analytics, Predictive Analysis, Prescriptive Analysis		
		# Self-Learning: LinkedIn Analysis, Netflix Analysis, Cricket and FIFA Analysis.		
2		Spatial Data Analysis	10	
	2.1	Contents and characteristics of Spatial data, Spatial data formats, spatial data bases		
	2.2	Introduction, Definition of GIS, Evolution of GIS , components of GIS Spatial Association rule mining, spatial hierarchal clustering, Set based decision and knowledge recovery		CO2 CO1
	2.3	Case study: GIS application for Spatial data mining using open source software tools		
		# Self-learning: QGIS, Hadoop, GeoSpark R PostgreSQL, PostGIS, Python		
3		Graph Analysis	10	
	3.1	Introduction to the Social Network, Clustering of Social-Network Graphs, Direct Discovery of Communities		
	3.2	Partitioning of Graphs, Finding Overlapping Communities, PageRank, Topic sensitive PageRank, HITS algorithm Simrank, Counting Triangles, Neighborhood Properties of Graphs		CO3
	3.3	# Self-learning: GraphX tools of Apache.		
4		Time series Analysis for prediction and forecasting	10	
	4.1	Introduction, Finding and Wrangling Time Series Data, Exploratory Data Analysis for Time Series, Simulating Time Series Data, Storing Temporal Data,		
	4.2	Statistical Models for Time Series, State Space Models for Time Series, forecasting methods, Testing for randomness, Regression based trend model :AR,MA,ARIMA, random walk model, moving average forecast, exponential smoothing forecast, seasonal models,		CO4
5		Data Analysis in Health Care Application	10	CO5
	5.1	Introduction, Components of EHR, Benefits of EHR-		
	5.2	Natural Language Processing and Analysis Of Clinical Text : Introduction , report analyser, text analyser, Core NLP Components Morphological Analysis , Lexical Analysis , Syntactic Analysis ,Semantic Analysis , Data Encoding .		
	5.3	Mining Information from Clinical Text: Rule-Based Approaches. Pattern-Based Algorithms ,Machine Learning		

Module No.	Unit No.	Details	Hrs.	CO
		Algorithms		
		Self-learning: Introduction to Social media Analysis for healthcare,		
Total			45	

Students should prepare all Self Learning topics on their own. Self-learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in IA and Laboratory Experiments.

Reference Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Michael J. de Smith, Michael F. Goodchild and Paul A. Longley	Geospatial Analysis: A Comprehensive Guide to Principles, Techniques, and Software Tools,	Wiley, Second Edition	2019
2.	Anil Maheshwari	<i>Data Analysis</i>	Mc Graw Hill	2017
3.	James, G., Witten, D., Hastie, T., Tibshirani, R.	<i>An introduction to statistical learning with applications in R</i>	Springer	2013
4.	Chandan K. Reddy and Charu C Aggarwal	<i>Healthcare data Analysis</i>	Taylor & Francis	2015
5.	U. Dinesh Kumar	<i>Business Analysis</i>	Wiley	2017
6.	Li, Deren., Wang, Shuliang., Li, Deyi	Spatial Data Mining: Theory and Application.	Spatial Data Mining: Theory and Application.	2016
7.	Albright and Winston	Bussiness Analysis	Cengage Publication	5 th edition, 2015
8.	Aileen Nielsen	Practical Time Series Analysis	O'Reilly Media, Inc.	1 st edition October 2019

*In addition to printed books, faculty can suggest (authentic) urls or e-books, e-contents etc.

Course Code	Name of the Course				
216H03L501	Data Analysis Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	-	02	-	02	
Credits Assigned	-	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	50	—	—	—	50

Course Objectives

- Comprehend methodologies of data analysis

Course Outcomes

At the end of the course students will be able to

CO 1	Understand basic concepts of data analytics to solve real world case studies	
CO 2	Apply the data analysis on geospatial system	
CO 3	Perform the graph data analysis	
CO 4	Perform Time series Analysis	
CO5	Apply the text data analytics in the field of Health care	

Term-Work:

Term work will consist of minimum 8 experiments/ tutorials covering entire syllabus of the course 'Data Analysis Laboratory'. Students will be graded based on continuous assessment of their term work.

Course Code	Name of the Course				
216H03C601	Machine Learning				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	03	-	-	03	
Credits Assigned	03	-	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	-	20	30	50	100

Course prerequisites: Basic data science concepts, Basics of programming in Python.

Course Objectives:

1. To establish a strong foundation in machine learning fundamentals.
2. To enable students to apply supervised learning techniques to solve real-world problems.
3. To introduce students to unsupervised learning methods for data exploration and pattern discovery.
4. To provide a foundational understanding of deep learning and artificial neural networks.
5. To introduce students to the concepts and principles of reinforcement learning.

Course Outcomes (CO):

After completing the course, student will be able to

CO 1	Recall and understand fundamental machine learning concepts,
CO 2	Apply supervised learning techniques to analyze and evaluate real-world applications.
CO 3	Employ unsupervised learning methods to explore, categorize data and differentiate patterns within datasets.
CO 4	Understand & describe mathematical foundation behind deep learning architectures.
CO 5	Gain a basic understanding of Reinforcement Learning (RL) principles and its potential applications.

Module No.	Unit No.	Contents	No. of Hrs.	CO
1	Machine learning foundation:		06	CO1
		What is Machine Learning? Types of learning, applications, Bias, variance, overfitting, under-fitting, cross validation and feature engineering.		
2	Supervised Learning		10	CO2
		Non Linear Regression, Multivariable Linear Regression, gradient descent learning algorithm and its variations & Ridge Regression and Lasso Regression (Regularization). Logistic regression for binary classification, Maximum Likelihood Estimation (MLE), L1 and L2 regularization for logistic regression & hyper-parameter tuning		
		Decision Trees: Structure of decision trees, Tree building algorithms (e.g., ID3, CART, C4.5), Ensemble learning, bagging and Random Forests.		
		Support Vector Machines: SVM for classification and regression, Margin and decision boundary, Soft margin vs. hard margin SVM, Kernel functions (e.g., linear, polynomial, radial basis function) & SVM for non-linear data separation		
3	Unsupervised Learning		10	CO3
	3.1	Dimension reduction using Principle Component Analysis		
	3.2	Hierarchical clustering, Agglomerative vs. divisive clustering & Density-based clustering (DBSCAN)		
	3.3	Introduction to Gaussian Mixture Models, Expectation-Maximization (EM) algorithm for clustering.		
	3.4	Performance Metrics (Accuracy, Precision, Recall, F1-score)		
4	Introduction to Deep learning		10	CO4
	4.1	Introduction to Artificial Neural Networks (ANN), Feed-forward and Back-propagation.		
	4.2	Activation Functions (ReLU, Sigmoid, Tanh & Softmax)		
	4.3	Introduction to Deep Learning Architectures: CNN		
5	Reinforcement learning and applications		09	CO5
	5.1	What is Reinforcement Learning? Key components: Agent, Environment, Reward, Policy, Value Function		
	5.2	Q-Learning		
Total			45	--

Reference Books

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/ Year
1	Tom M.Mitchell	Machine Learning	McGraw Hill	2017
2	M. Gopal	Applied Machine Learning	McGraw Hill	2018
3	Ian Goodfellow, Yoshua Bengio, Aaron Courville	Deep Learning	An MIT Press book	2016
4	Aurélien Géron	Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd Edition	O'Reilly	2019
5	Sebastian Raschka Vahid Mirjalili	Python Machine Learning, Second Edition: Machine Learning and Deep Learning with Python, scikit-learn, and TensorFlow	Packt	2017

Course Code	Name of the Course				
216H03L601	Machine Learning Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	-	02	-	02	
Credits Assigned	-	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
		50	-	-	-

Laboratory experiments will be based on the entire syllabus of the course 216H03C601, 'Machine Learning'. Students will be graded based on continuous assessment during laboratory.

Course Code	Name of the Course				
216H03L602	Data Visualization Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	-	02	02	
Credits Assigned	--	-	02	02	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

***Onscreen examination**

Course prerequisites (if any):

Basics of statistics, database and data analysis

Course Objectives

- Employ best practices in data visualization to develop charts, maps, tables, and other visual representations of data
- Use visualization tools to conduct data analysis, especially exploration of an unfamiliar dataset.
- Create compelling, interactive dashboards to combine several visualizations into a cohesive and functional whole.
- Use data visualizations, dashboards and Stories to support relevant communication for diverse audiences.

Course Outcomes

At the end of successful completion of the course the student will be able to

CO1	Learn how to locate and download datasets, extract insights from that data and present their findings in a variety of different formats
CO2	Detect and understand the stories within datasets and its applications.
CO3	Apply data visualization best practices
CO4	Design static charts, interactive Dashboards and data stories

Module No.	Unit No.	Details	Hrs.	CO
1	Introduction Data Visualization		05	CO 1 CO 2
	1.1	Introduction to data visualization and its need. Data analysis lifecycle. A Visual Revolution, Various types of visualization with its best practices. From Visualization to Visual Data Storytelling: An Evolution, From Visual to Story: Bridging the Gap		
	1.2	Data Fundamentals, Collecting data, Preparing Data		
	1.3	Gestalt Design principles		
	1.4	Data cleaning using Excel		
2	Visualization Foundations		07	CO 3
	2.1	Induction to Data foundation		
	2.2	Addressing the design of visualizations Human Perception and Information Processing		
	2.3	Geospatial Displays : Connecting to Geographic Data Assigning Geographic Roles Creating Geographic Hierarchies Proportional Symbol Maps, Choropleth Map		
3	Visualization Techniques		08	CO3
	3.1	Visualization Techniques for Spatial Data, implicit or explicit spatial or spatiotemporal attribute, mapping of the data attributes to graphical attributes		
	3.2	Visualization Techniques for Time-Oriented Data, Handling the temporal dimension. time and time-oriented data. TimeBench, visual analytics of time-oriented data.		
	3.3	Visualization Techniques for Multivariate Data		
	3.4	Visualization Techniques for Trees, Graphs, and Networks		
	3.5	Text and Document Visualization		
4	Exploratory Visualization		05	CO3
	4.1	Data Joins Best Practices o Sorting, Top N, bottom N ,Filtering , Maps		
	4.2	Visual Analytics		
	4.3	Optimal visualization types , Binning values, Calculated fields , Table calculations , Level of Detail calculations		
5	Storytelling and Dashboards		05	CO4

Mod ule No.	Unit No.	Details	H rs.	CO
	5.1	Storytelling Multivariate displays The Science of Storytelling The Power of Stories Context in action Exploratory versus Explanatory Analysis Structuring Stories Audience Analysis for Storytelling Steps to Visual Data Storytelling The Important Role of Feedback		
	5.2	Create dashboard Working with dashboard Publishing through dashboard		
	5.3	Design principles for creating effective dashboards Implementing interactivity and storytelling elements Presenting insights effectively using Tableau and Power BI features		
		Total	30	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Lindy Ryan	<i>Visual Data Storytelling with Tableau</i>	Pearson Education	First edition, 2018
2	Cole Nussbaumer Knaflic	Storytelling with Data	Wiley	First edition, 2015
3	Alberto Ferrari and Marco Russo	Introducing Microsoft Power BI	Microsoft Press	2016
4	Nisal Mihiranga	Power Bi Data Modelling	BPB	First Edition 2022
5	Brett Powell	Mastering Microsoft Power BI	Packt	First Edition , 2018

Course Code	Course Title							
116h54C701	Advanced Data mining							
	TH		P	TUT			Total	
Teaching Scheme (Hrs.)	03		--	--			03	
Credits Assigned	03		--	--			03	
Examination Scheme	Marks							
	CA		ESE	TW	O	P	P&O	Total
	ISE	IA						
	30	20	50	--	--	--	--	100

Course prerequisites: Data mining

Course Objectives:

1. To analyze various algorithms and techniques to mine complex data beyond conventional record data to mining complex structure and complex data
2. Extract useful knowledge from massive data sources using distributed computing solutions for data intensive applications
3. To develop the basic skills necessary to pursue research in data mining.

Course Outcomes

On completion of the course students will be expected to

1. Design and implement frequent itemset mining and clustering algorithms to discover patterns and groupings in large transactional and streaming datasets using memory-efficient techniques.
2. Apply stream mining techniques to analyze and summarize continuous data using efficient algorithms for sampling, filtering, and counting.
3. Apply tree mining frameworks to extract structured patterns from complex hierarchical data.
4. Apply text mining and recommendation techniques using vector space models, classification algorithms, and dimensionality reduction methods.
5. Apply sequence and time series mining techniques to discover patterns and trends in multivariate temporal datasets.

Module No.	Unit No.	Details	Hrs.	CO
Advanced Data Mining				
1	Frequent Itemset and Clustering		03	CO1
	Introduction to Data mining, KDD, The Market-Basket Model, Market Baskets and the A-Priori Algorithm, Handling Large Datasets in Main Memory, Limited-Pass Algorithms, Counting Frequent Items in a Stream Clustering: The CURE Algorithm, Clustering in Non-Euclidean Spaces, Clustering for Streams and Parallelism.			
2	Mining Data Streams		06	CO2
	The Stream Data Model, Sampling Data in a Stream, Filtering Streams, Counting Distinct Elements in a Stream, Estimating Moments, Counting Ones in a Window, Decaying Window			
3	Mining complex structures		10	CO3
	Mining trees- Tree Model Guided Framework, TMG framework for mining ordered & unordered subtrees, Tree Mining Applications, Mining maximal and closed frequent trees, Tree mining application			
4	Text mining and recommendation system		08	CO4
	The vector space model: Parametric and zone indexes, Term frequency and weighting, The vector space model for scoring Variant tf-idf functions The text classification problem, Naive Bayes text classification The Bernoulli model, Properties of Naive Bayes, Feature selection, Evaluation of text classification Document representations and measures of relatedness in Vector spaces, Classification with more than two classes Recommendation Systems: Content-Based Recommendations, Collaborative Filtering Dimensionality Reduction			
5	Sequence mining & Multivariate and Time series mining		08	CO5
	Sequence Mining- Characteristics of Sequence Data, Problem Modelling, Sequential Pattern Discovery, Timing Constraints Applications in Bioinformatics Multivariate and Time series mining- Importance of Multivariate and Time series data, Sources of MVTs data, Mining MVTs data: Sign Language Data, Agro-meteorological Data, Generalized sequential patterns algorithm, DTW algorithm			

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Hadzic F., Tan H. & Dillon T. S	Mining data with Complex Structures	Springer	2011
2.	Yates R. B. and Neto B. R	Modern Information Retrieval	Pearson Education	2005
3.	Han J. & Kamber M	Data Mining: Concepts and Techniques	Morgan Kaufmann Publishers	Third edition, 2011
4.	Christopher D.M., Prabhakar R. & Hinrich S	Introduction to Information Retrieval”	Cambridge UP	Online edition, 2009
5.	Tan P. N., Steinbach M & Kumar V	Introduction to Data Mining	Pearson Education	2006

Course Code	Name of the Course				
216H03L701	Advanced Data Mining Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	-	02	-	02	
Credits Assigned	-	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	50	-	-	-	50

Laboratory experiments will be based on the entire syllabus of the course 216H03C701, 'Advanced Data mining'. Students will be graded based on continuous assessment during laboratory.



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